UFOs or Rocks? A Dive Into the True Nature of UFO Sightings vs. Close-Earth Object Approaches

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Introduction

Every year, countless UFO sightings ignite curiosity, debate, and speculation. But could the explanation be less otherworldly than we imagine? The UFOs or Rocks Team,alongside the MythBusters, delves into this question, exploring if many of these sightings correspond with the trajectories of close-Earth objects (NEOs). By comparing UFO reports with NEO data, we aim to uncover patterns and offer evidence-based insights. Dive with us into this data-driven analysis to determine: Are we witnessing extraterrestrial phenomena or merely celestial bodies passing by and are the patterns in UFO sightings truly novel or just predictable?

Hypothesis

We hypothesize that:

H1(null): UFO sightings cannot be explained solely by the presence of near-Earth object (NEO) approaches.
H1(alternative): UFO sightings can be explained solely by the presence of near-Earth object (NEO) approaches.

- Higher frequency of NEO approaches should reflect a higher number of UFO sightings.
- Times in which NEOs are bigger should reflect a higher number of UFO sightings.
- Times in which NEOs are closer to Earth in distance should reflect higher number of UFO sightings

H2(null): Patterns within UFO Sightings are not predictable and are novel.
H2(alternative): Patterns within UFO Sightings are predictable and not particularly novel.

Data Source and Description

Primary - National UFO Reporting Center Online Database

- Our primary data source came from the National UFO Reporting Center Online Database, which we scraped using Python. This site allows individuals to report sightings that may resemble UFOs.Within the data, we found the following details:
  - Date and time of sighting
  - Location, including city, state, and country
  - Description of the UFO's shape
  - Duration of the sighting
  - Free-form text summarizing the sighting
  - Upload date of the report
  - Indication if there were any associated images
  - We scraped 144088 records, covering the period from 1400 to 2023. We scraped by month/year report titles because of an issue with date formatting. Specifically, the “Date/Time” column in the data only showed “23” for the year, making it

unclear if it referred to 1923, 1823, 2023, etc. By scraping by month/year, we were able to correct the year in the Date/Time column using the full year from the month/year report titles.

<u>Secondary - NASA'S Center for Near Earth Object Studies (CNEOS)</u>

- For an understanding of close-Earth celestial occurrences, we tapped into the database provided by NASA's Center for Near Earth Object Studies. We used their API due to the limitations inherent in their CSV data offerings. The API was a perfect fit to get all the data they had from 1900-2023.Our method involved:
  - Sending a GET request specifying our desired date range.
  - Receiving the data in JSON format.
  - Converting it to a pandas dataframe.
  - This API-fetched dataset largely mirrors the content available on NASA's website, albeit with a few discrepancies. While column names varied, their content was consistent. However, two points of difference emerged:
    - The API dataset lacked a "Rarity" column.
    - The "Diameter" column from the API proved to be sparser than its website counterpart.
  - We scraped all 23616 records on the website at the time.
  - The data is comprehensive, detailing the trajectories and characteristics of asteroids and comets as they skim past Earth. Key data fields encompass the timing of their approach, distance covered, velocity, and magnitude.

Data Cleaning

<u>National UFO Reporting Center Online Database – UFO data</u>

- Since we are interested in UFO reports within the United States of America, we created two dataframes, one for 'USA' and the rest in another. In order to double check that the contributors did not omit the 'Country' column or misspelled, we checked the non USA dataframe for unique 'Country' names in order to see if there were any variations of 'USA'. Each of the variations were checked to see if they could be appended to the USA only dataframe. This reduced the dataframe into 128,793 rows.
- Next, we moved on to check the 'State' column to see if all of the States were properly abbreviated and for any inconsistencies. There were several misspelled O for the number zero and several non USA states listed. After looking at the individual rows, relevant data was edited to reflect the correct states and taken out of the USA only dataframe. We also checked for nulls and they were dropped since 'State' is an important component to our exploratory questions. There were 12 rows with nulls. Once all of this was complete, the data frame was sorted by the correct alphabetical order of full USA state names.
- Shapes' column was capitalized to join any input that declared the same shape. To ensure distinct shapes, the inputs of the similar shape such as 'Circle' and 'Orb' into 'Sphere', were merged. All nulls were turned into 'Unknown.'
- The 'Date/Time' column was split into two 'Date' and 'Time' to match the NEO dataset format. Any remaining nulls for Date and/or Time were dropped.
- A new column called "Duration_min" was created to convert all the durations reported to minutes. Initially, textual representations of numbers (e.g. "one", "two") were translated to their numeric equivalents if they weren't already given in numeric form. Following this, durations specified in hours or seconds were standardized to minute equivalents. If a range was provided (e.g. 10-15 mins), then the average was taken. Unrecognized formats like "ongoing" or "unsure" were dropped.

- We dropped the two columns we did not need, the date the report was posted and whether there was an image attached with the report or not. We also dropped the years before 2000 in order to have more reliability in the reports and ended up with a final cleaned UFO dataset of **110,607 rows x 8 columns.**

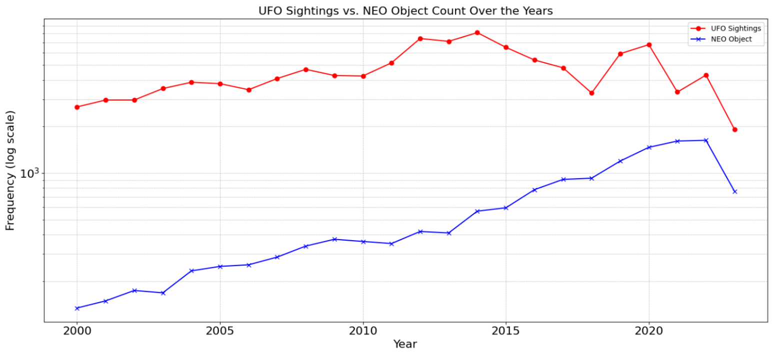
NEO Earth Close Approaches – Near-Earth Approach Object Data

- We downloaded the dataset through an API call as the site did not allow downloads of data of more than 10,000 rows. Our initial dataset ranged from January 4, 1900, to August 1, 2023. We called this DataFrame "nasa". In total, the original DataFrame had 23,616 rows.
- We noticed that data downloaded via the API call lacked diameter data, with the column comprising 97.18% of "NaN" data. As a result, we decided to remove the diameter column to look towards absolute magnitude H to indicate NEO sizes.
- We further renamed the columns into more understandable and descriptive names.
- We then separated the datetime column into two: 'date' and 'time'. Initially, the two data points were combined into a single column that had an alphabetical structure of date (e.g. 21-Jul-1997) and time that included seconds (e.g. 00:21:19). In order to translate the column into a datetime object in Pandas, we separated the date portion as its own column and converted it into a standard DataFrame datetime object of yyyy-mm-dd. Similarly, we created a separate column for time to restructure it into the hh:mm DataFrame datetime format. We conducted this separation using a string operation of the origin column.
- We then removed columns that do not contribute to our analysis of UFO sightings and NEO approaches, such as orbit class, kind of NEOs, and NASA designations of NEOs.
- We collectively agreed to look at data starting from January 1, 2000, to limit our scope to the 21st century. Thus, we created a separate DataFrame called "nasa_data" with the date range from 2000-01-01 to 2023-08-01. This new DataFrame contained **14,344 rows x 12 columns**.

UFO Sightings and NEO Approaches: A Temporal Analysis with Insights into NEO Exploration

By comparing directly the number of UFO sightings with the number of NEO approaches throughout the years 2000-2023 in figure 1, we initially see a similar increasing trend between the two sightings at least until 2015. After 2015, while NEO sightings still increased, no doubt because of advancements in technology, UFO sightings seem to be a bit more sporadic. This may be because UFO sightings rely on reports from the general public rather than on sophisticated equipment.

Figure 1.



As we investigate deeper, we begin to see more specific insights between NEO approaches and UFO sightings. According to figure 2, NEOs are bigger in the summer but much farther away than spring and fall seasons.

Figure 2.

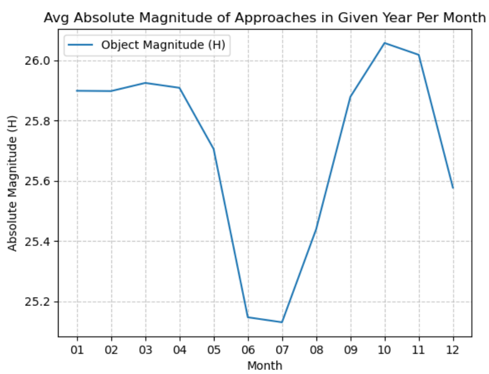
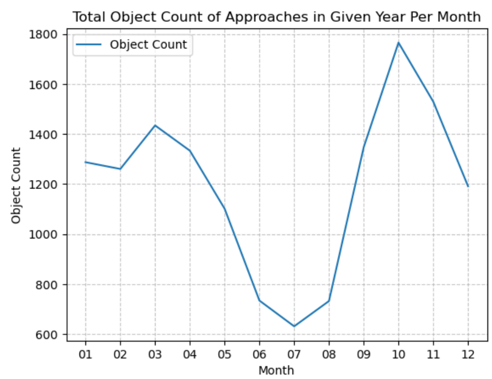
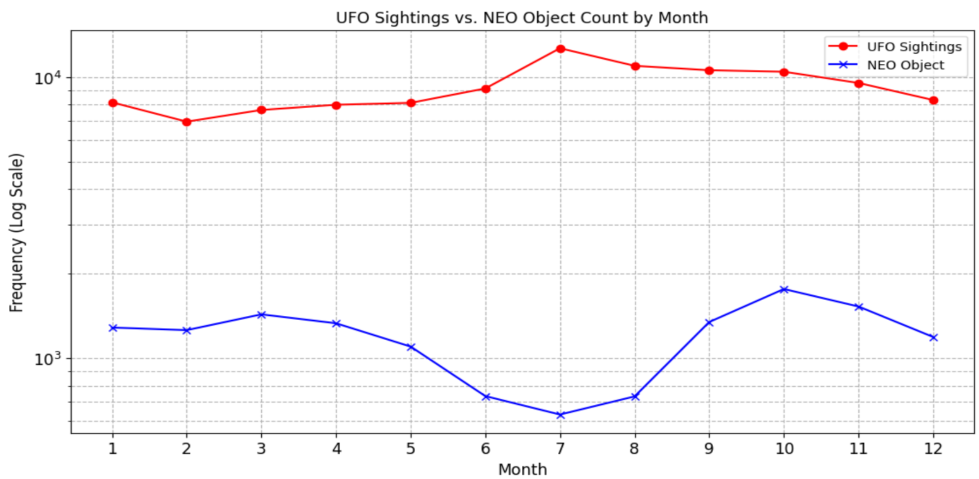


Figure 3.



Absolute magnitude (H) and asteroid diameter have an inverse relationship: The smaller H, the bigger the diameter. H appears to be the smallest on average in mid-summer months (June and July) of a given year. Interestingly, this reflects with the number of UFO sightings monthly, in which sightings tend to increase in the months of July, August and September (figure 4). Since June and July had fewer yet much bigger NEOs, this may explain the simultaneous increase in UFO sightings in these months as well (figure 4). However, according to figure 4, NEO approaches to Earth also drop the lowest in summer months. While we can theorize that bigger objects provide better visibility, we cannot deduce from this alone if this actually allows for correlated UFO sightings in the face of decreasing approaches.

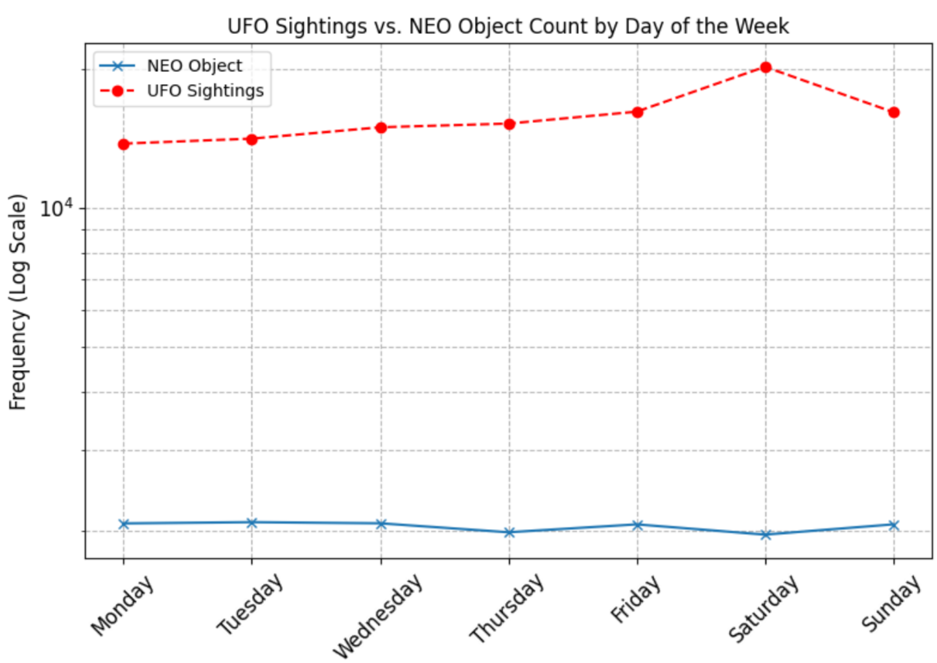
Figure 4.



Interestingly, as shown in figure 4, NEO approaches appear to have an inverse relationship with UFO sightings on a monthly basis for years 2000-2023. This is likely due to the seasonal nature of NEO approaches, with decreased yet larger sightings in the summer months. The increase in sizes of NEOs may contribute to better visibility and increased sightings as UFOs. The UFO sightings increasing in the summer/fall time may be because people are out more during those months.

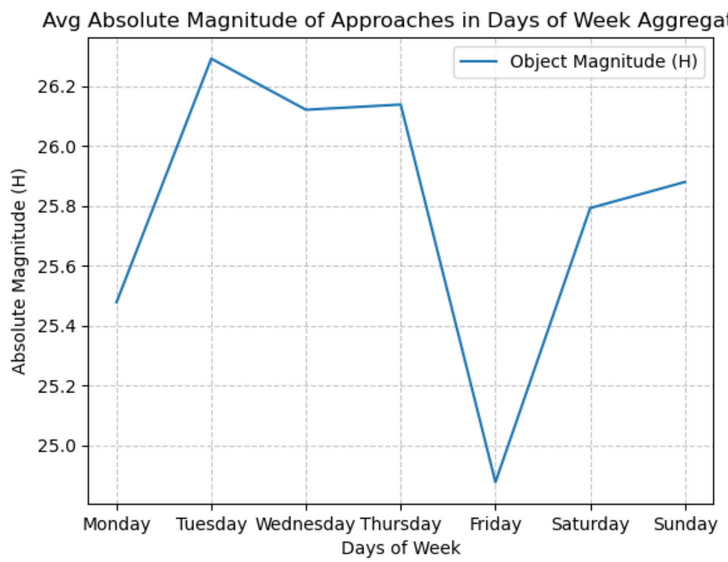
We further investigated more granular counts of NEO approaches by days of the week.

Figure 5.



In figure 5, we can see that days earlier in the week (Sundays to Tuesdays) have higher total counts of NEO approaches. Interestingly, UFO sightings tend to spike on Saturdays, a direct inverse with that of NEO approaches. This graph also supports the lack of positive correlation between UFO sightings and NEO approaches. However, the increase in Saturday sightings in UFOs may be correlated to the large increase of NEO sizes (decrease in H) on Fridays, shown in figure 6, resulting in possibly better visibility on early Saturday mornings. Furthermore, the uptick in UFO sightings on Saturdays could be attributed to the higher outdoor activity of people during weekends.

Figure 6.



Examining figure 7 and 8 suggests there is no clear evidence that NEO distance correlates with frequency of UFO sightings. In both monthly and weekly views of NEO distances in kilometers, close proximities do not corroborate with hypothesized higher numbers of UFO sightings.NEOs are farthest from Earth in summer months and on Friday, which do not coincide with increases in UFO sightings in summer months and on Saturdays.

Figure 7.

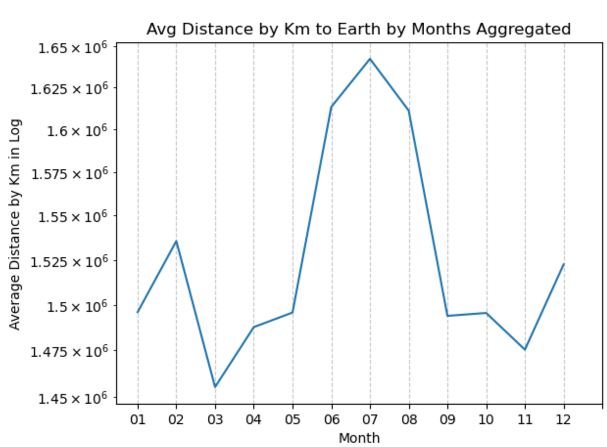
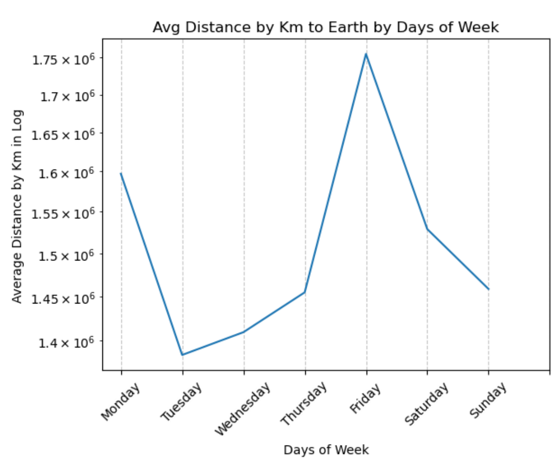
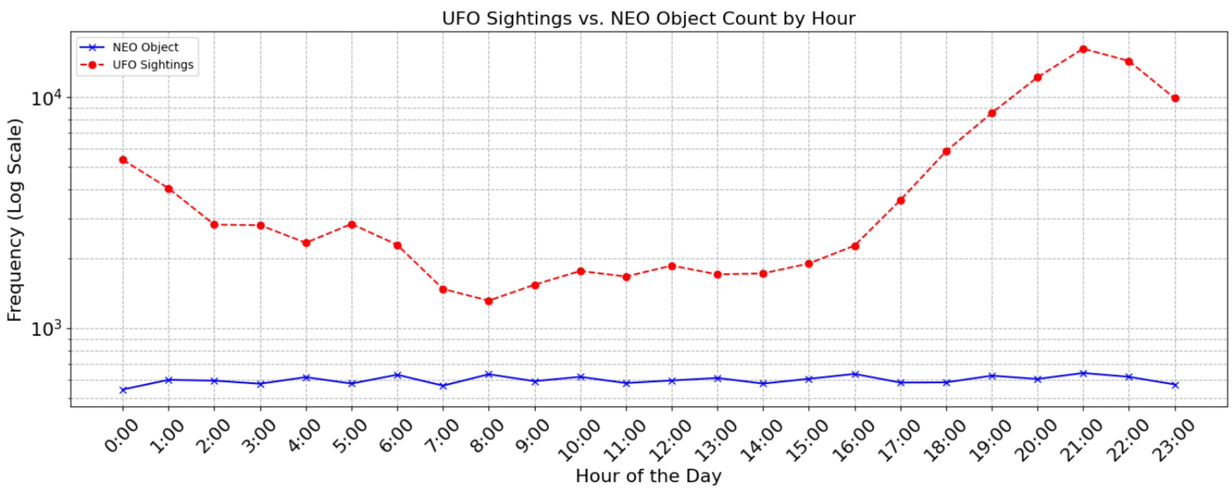


Figure 8.



Delving deeper, our assessment of hourly patterns in UFO and NEO sightings in figure 9 revealed consistent NEO sightings across the day, courtesy of continuous technological monitoring. UFO sightings, being contingent upon human observers, seem to coincide with typical outdoor hours between 4p to 9pm. Nevertheless, this hourly distribution doesn't present a parallel for both sightings.

Figure 9.

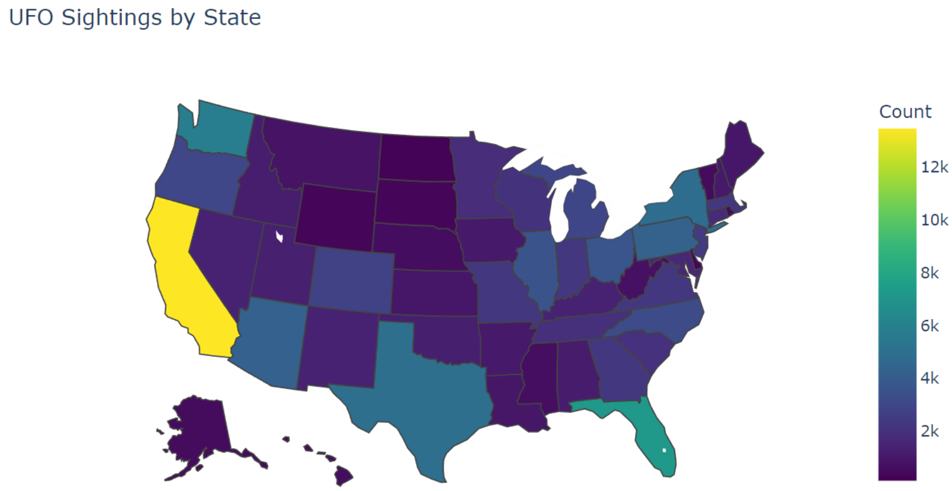


In conclusion, there are multiple inconsistencies in the patterns. Specifically, despite larger NEOs being visible in summer, which is theorized to contribute to more UFO sightings, the decrease in NEO approaches during these months contradicts this theory. Moreover, no clear evidence suggests that NEO distance from Earth correlates with the frequency of UFO sightings. The hourly distribution also doesn't show a parallel between both sightings. Given these observations, it's not conclusive that UFO sightings can be solely explained by the presence of near-Earth object (NEO) approaches, reducing support for our alternative hypothesis (H1).

UFO Geographical Variation

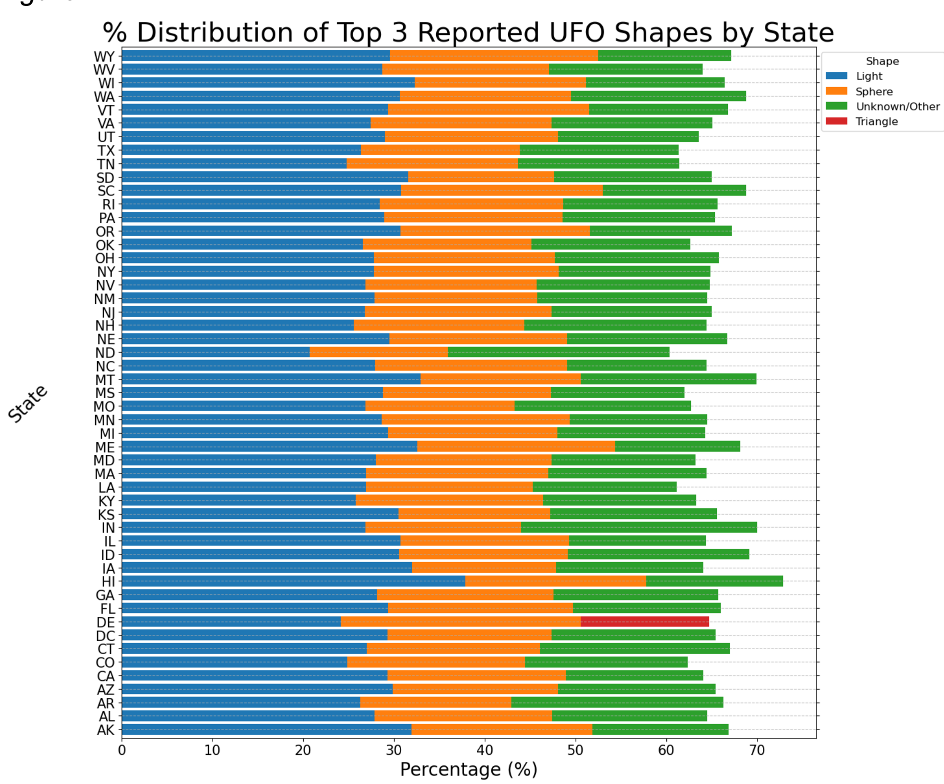
In addition to the expected patterns like higher UFO sightings at night and during summer, we further explored distinct patterns by geographical variation. According to figure 10, when observing the UFO sightings by State, sightings in California exceed all others. Given that California boasts the highest population in the United States, the elevated count in this state comes as a little surprise. Similarly, states with the next highest counts, such as Texas, Florida, and New York, align with their significant population size.

Figure 10.



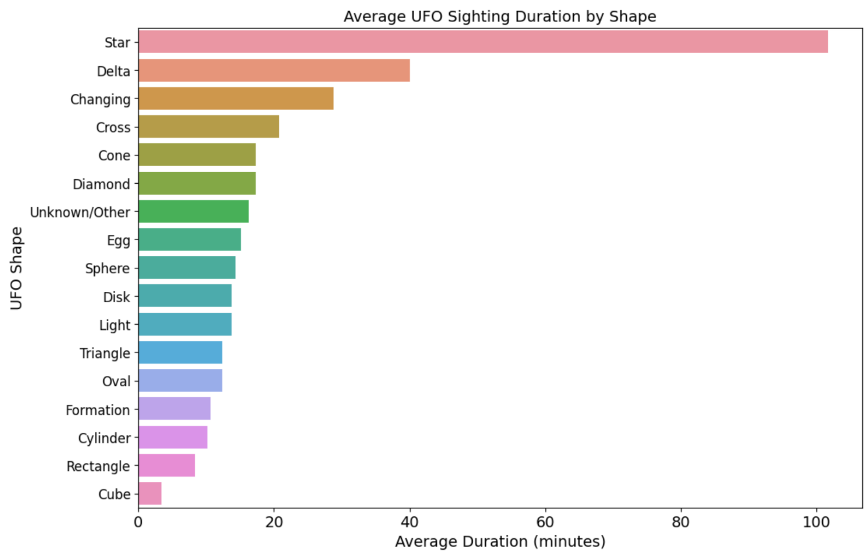
Shapes are another component of the UFO sightings to observe. According to Figure 11, notably three shapes - light, sphere and unknown/other - stand out across all states except Delaware. These prevalent shapes align with our common perceptions of UFOs, strengthening the support for our alternative hypothesis (H2). In a galaxy this vast, if UFO sightings were to become common, it's improbable that we would consistently encounter these specific patterns perpetuated by media portrayals of UFOs.

Figure 11.



We also analyzed duration in minutes for all the shapes reported. From figure 12, it seems like people reported seeing the shape “Star” for a longer duration than the other shapes. As we saw from figure 9, since people mostly sight UFO like objects at night and stars also do come out at night, this can be indicative of a star shaped UFO object, or an actual star.

Figure 12.



UFO Description

We analyzed all the words in the descriptions used to describe UFO sightings and made a word cloud represented by figure 13. The bigger the word in the figure, the more people used it in describing what they saw. The words seem very vague and predictable. A lot of people describe seeing a bright light object moving in the sky, but it is hard to glean any novel information from the descriptions.

Figure 13.
Word Cloud of UFO Description

Conclusion

In conclusion, the analysis presented here sheds some light on the complexities of interpreting UFO sightings and Near-Earth-Objects (NEO) data. Several important caveats should be acknowledged: inclusion of the same observations made by several individuals, significant variation in duration values (“ongoing” or “unsure” or “1:00”) and potential influence of altered states due to intoxication.

For hypothesis H1, in absence of substantial evidence to support the alternate hypothesis, it is reasonable to favor the null hypothesis. The connection between UFO sightings and Near-Earth-Object approaches lacks clear validations and requires further analysis on the Mythbusters end.

Conversely, the identifiable patterns observed in many UFO sightings seem to follow predictable trends, potentially confusing common celestial activities, supporting the H2 alternate hypothesis. This can be observed by the common shapes and keywords (star, light, bright) used to describe these sightings, emphasizing a potential need for more sophisticated analytical methods when investigating UFO sightings. It is necessary to acknowledge the constraints of the study, highlighting the need for further investigations into UFO phenomena and NEO observations. We, the UFO or Rocks team with the Mythbusters, will be back next time.